



Lady Shri Ram College for Women

(University of Delhi)

Department of Statistics

**Incidence of Multimorbidity and Its Related Factors Prevalent
Among Ageing Population in India: An in-depth Analysis
Based on The Longitudinal Ageing Study of India
(LASI) Data
An Academic Project**

Submitted in partial fulfilment of the requirements for the award of the degree of

**Bachelor of Science (Honours)
with Research
in
Statistics**

Submitted by
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Under the supervision of
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CERTIFICATE

Certified that the Academic Project entitled **Incidence of Multimorbidity and Its Related Factors Prevalent Among Ageing Population in India: An in-depth Analysis Based on The Longitudinal Ageing Study of India (LASI) Data** submitted by Rekha from the Department of Statistics, Lady Shri Ram College for Women, University of Delhi for the award of the degree of Bachelors in Science (Honours) with Research in Statistics is an original piece of work conducted under my supervision. The percentages of plagiarism and AI-generated content in this work have been assessed, and they are within acceptable limits.

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This Academic Project submitted in the academic year 2025-26 for the award of the degree of Bachelors in Science (Honours) with Research in Statistics has not been submitted in part or in full for the award of any degree or diploma to any other university. The percentages of plagiarism and AI-generated content in this work have been assessed, and they are within acceptable limits.

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1. INTRODUCTION

Multimorbidity, majorly defined as the presence of two or more chronic diseases in one person at the same time. It became a major public health problem worldwide. Survival rate has increased as health services and living conditions are improving. This shift directly affects demographic transition, which means that when a country grows due to socioeconomic conditions, birth rate and death rate become low, and the patterns of the society's age structure become different. Early children's rate was higher than older, but now it is reversing. While increasing the rate of older adults proves that there has been great improvement in public health, nutrition, and medical technology, it has also created many challenges for health services, jobs, social security systems, and support systems between different generations. Around the world, population aging is increasing very fast in a way never seen before. As stated by the expected population in future, the number of persons aged 60 years and above will be greater than the number of children below the age of 14 for the first time in future. Better life expectancy has helped cause this change. From the past many years, global life average age of living at birth has increased to a great extent for both men and women, while fertility rates have declined sharply from historically high levels to near replacement levels (Birth rates are close to the level needed to keep the population size stable (around 2.1 children per woman), without growing or shrinking much.) in many regions. These trends have resulted in a growing share of older people and a gradual decrease in younger age cohorts. A notable feature of global population aging is its increasing concentration in developing regions. Although aging was initially associated with high-income countries, low- and middle-income nations are now experiencing faster aging transitions, often within much shorter time spans and without proper systems. The majority of the world's elderly population is presently located in developing areas, and this proportion is projected to increase further in the coming decades. Thus, many countries face the challenge of aging before not fully developed economically, health services are under pressure, and family support.

Systems, and government help schemes. Population aging also brings changes in dependency relationships. In the beginning, fewer children mean less dependency, but over time, more elderly people increase it again. Additionally, Older populations have more women, because women usually live longer than men, resulting in a higher proportion of Older women who usually have more financial problems and health issues. These worldwide trends show the

urgent need for accurate and consistent, and longitudinal Data to study the many aspects of aging and help make good policies.

In the past years, previous research papers suggested that multimorbidity is not only present in high-income countries, but it is becoming common in low- and middle-income countries as well. Countries like India and China, **Pati S, Swain S, Hussain MA, et al (2023)** found that these countries hold 36% population of the world's population, and are aging at a high rate. A systematic review by Zhang et al. (2022) observed a high 30.7% prevalence of multimorbidity in these countries, and hypertension was found to be as most common disease, with combinations of chronic diseases. Past reviews, such as Violan et al. (2014), searched the increasing commonality of multimorbidity in the world, but patterns of co-occurring diseases and evidence from Asian countries remained limited.

In the context of India, researchers commonly use nationally representative studies' data from the Longitudinal Aging Study in India (LASI) have shown important insights into the burden of multimorbidity. Many studies have identified that multimorbidity is high among older adults and is strongly positively related with age, being female, urban areas, and higher socioeconomic status confirmed by Chauhan et al. (2022) and Patel et al. (2023). Sinha et al. (2022) verified that in different states of India burden of multimorbidity was also different because of health care services, availability of hospitals, and different lifestyles. The prevalence of multimorbidity was more common in urban areas; as a result, they often go hospital and increase the expenditure from their pockets.

Brahmankar et al. (2024) established in addition to physical health, more than one chronic disease also affects mental health and reduces the quality of life of an individual. The studies that were done in India clearly indicate that in older adults and in the middle-aged group, multimorbidity was highly linked to depression. And Singh et al. (2025) explored that an increase in the number of diseases is associated with a decrease in the quality of life. Debsarma et al. (2022) analysed particularly in women, multimorbidity was high due to social, economic challenges and biological factors in their life expectancy.

Although more research is being done in this area, many studies only include certain age groups, give only a few results, or use data based on a one-time observation. Therefore, Future studies should examine how multiple long-term illnesses affect people, along with things like income, emotional well-being, and daily habits. Completing the missing parts

could lead to better healthcare setups that handle several illnesses at once in aging populations, rather than targeting just a single ongoing illness.

1.1 IDENTIFICATION OF THE RESEARCH PROBLEM:

Our primary goal of research is based on the fact that, despite the gains in longevity, the rapid increase in the elderly population has not been accompanied by a proportional development of healthcare services, long-term care facilities, or comprehensive social security systems. Hence, our research area is to undertake the study of the multimorbidity of the elderly population of age ≥ 45 years in India and to identify the related covariates like age, income, sex, urban-rural setup, education, etc., which make a significant impact on the levels of multimorbidity.

1.2 LASI DATA

India represents a critical case for the study of population aging due to its large population size, rapid demographic transition, and significant regional diversity. Although India continues to have a relatively young population compared to many developed countries, the absolute number of older people is increasing rapidly. Recognising the need for comprehensive and policy-relevant evidence on aging, the Longitudinal Aging Study in India (LASI) was thought of as a nationally representative study focusing on the health, economic, and social well-being of India's aging population.

LASI is a large-scale, longitudinal survey conducted under the leadership of International Institute for Population Sciences, in collaboration with the Ministry of Health and Family Welfare, Harvard T. H. Chan School of Public Health, and University of Southern California. The study is aligned with the global Health and Retirement Study (HRS) family, enabling international comparability of aging-related research.

The first wave of LASI (Wave-1), conducted during 2017–2019, serves as the baseline for future follow-ups. It covers individuals aged 45 years and above and their spouses, irrespective of age, across all states and union territories of India. By adopting a life-course perspective, LASI captures transitions in health, work, income, family arrangements, and social participation as individuals move from mid-adulthood into older ages. The study was designed to generate evidence that can support planning and evaluation of health and social

policies targeted at older adults, particularly in the context of rising non-communicable diseases and changing family structures.

LASI stands out because it covers a wide range of topics in great detail. It integrates household-level and individual-level information with objective health measurements, making it one of the most comprehensive aging datasets available for India. The longitudinal design of the study allows researchers and policymakers to examine not only the current conditions but also changes over time, thereby strengthening the causal understanding of aging processes. For the study of aging population using lasi data, we have studied the following research paper in details.

1.3 Survey Instruments of the LASI Study

The study uses instruments designed to capture the various aspects of aging of the respondents. It has **multiple questionnaires and examination tools**, administered at both the household and individual levels.

The **household questionnaire** collected information on household composition, demographic characteristics, housing conditions, access to basic amenities, income sources, consumption, expenditure, assets, debts, and health insurance coverage. This section helps in understanding the living conditions and economic environment of the respondents.

The **individual questionnaire** is administered to all age-eligible respondents (45 years and above) and their spouses. It includes questions based on demographic background, education, marital history, migration, employment status, job characteristics, retirement, pension and social security, health behaviours, healthcare utilisation, and social participation. It also includes important questions based on chronic health conditions, functional limitations, and mental health outcomes, which gives information on ageing-related vulnerabilities.

An important feature of LASI is the integration of **health examination and biomarker instruments** alongside self-reported data. Various equipment and protocols were used by trained health investigators to conduct direct physical measurements. These instruments include sphygmomanometer for blood pressure, spirometers for lung function, weighing scales and stadiometers for anthropometric profile of respondents, hand dynamometers for grip strength, vision charts, and balance and walking tests to assess physical functioning.

Dried blood spot samples were also collected to obtain information on metabolic and cardiovascular health.

The survey also included **psychometric and cognitive tools** to measure mental health and cognitive functioning. Cognitive tests were used to assess memory, orientation, arithmetic ability, verbal fluency, and executive function. Depression is measured using screening instruments such as the Centre for Epidemiologic Studies Depression Scale and the Composite International Diagnostic Interview Short Form. To minimise the measurement error and ensure consistency, proper guidelines for the interviewer and detailed manuals for instruments were provided. To accommodate a diverse sample the questionnaires were administered in multiple regional languages.

1. Literature review

1.1 Examining the Impact of Disasters on Depression among Older Adults: Insights from the Longitudinal Aging Study in India (LASI), 2017-18

Shri et al. (2025) performed a study on the association between disaster exposure and depressive symptoms in older adults in India using nationally representative data from the Longitudinal Aging Study in India (LASI) 2017–18. According to Anand A et al. (2015), depression is a serious mental health problem among 45+ older adults in India. Tang B et al. (2014) verified that natural and human-made disasters like floods, droughts, earthquakes, fire, and road accidents were found to be most common in India, and these lead to physical injury, loss of property, migration, social isolation, and economic stress. All these factors increase the chances of depression in older adults. Looking into India, research by Becker AE and team in 2013 along with work from Dwivedi SN's group in 2024 focused on kids or broad population groups within certain areas - national data hardly ever came into play. When it comes to older people in India, still missing is any look at how both nature-caused and human-driven Catastrophes cause suffering to their minds. To fill this gap, the study was conducted. The main objective of the study was to determine the depression level in victims of disasters. Also, understand which socioeconomic factors were associated with depression.

The starting sample size was 72,000. In the analysis, a sample was 65,114 participants, 45+ older adults, and the outcome variable depression was calculated using the CES-D scale (a scale of 10 questions). about sadness, tiredness, feeling alone - those kinds of things. Ten signs checked, seven that point to struggle, three that show up when someone seems down.

Marked either zero for not there, or one if it shows. Add them all, get a number between zero and ten. When four or more appear, the label lights up if depression was noted. Below four Not marked, and {0,if CES-D score<4 . So the dependent variable is **binary**. Independent variables were taken from disaster exposures, and the control variables were age, sex, education, marital status, living arrangements, wealth index, etc., to estimate the net effect of disasters was estimated. Because the response variable was binary, they used the logistic regression model given as

$$\log\left(\frac{p(y=1)}{1-p(y=1)}\right) = \beta_0 + \beta_1 D + \sum \beta_i x_i$$

Where -Y = depressive symptoms, D = disaster exposure, Xi= control variables, β_0 = intercept, and β_i = regression coefficients. Propensity score matching (PSM) was performed to reduce the selection bias, and disaster exposure was taken as a treatment. It addressed the question of how much depression would change if disaster-affected individuals had not been harmed.

The major findings of the study were as follows: 26.9% of older adults were suffering from depressive symptoms, 4.1% reported health affected by disasters, and depression was found to be higher in females, rural residents, the poor, less educated, and multimorbidity individuals. From the regression, the results obtained that health affected by any disaster adjusted odds ratio was 1.48, health affected by both disasters AOR=2.73. This indicates disaster-affected older adults were 2.7 times more likely to be depressed. PSM calculated ATT = 0.097 shows that disaster exposure increases the chances of depressive symptoms by 9.7 per cent.

This research used cross-sectional data, self-reported disaster exposure, timing of disaster measured, and possible recall bias, which indicate the limitations of the study.

This study gave a conclusion that the disaster-related health impacts significantly increase depressive symptoms in older adults in India. From the statistical analysis, they observed that disasters were a strong factor associated with mental health in older adults, but the factor was not properly studied.

1.2 Aging, multimorbidity, and quality of life: a mediation analysis using a longitudinal aging study in India

Bharti Singh et al. (2025) aimed to obtain the measurements of QoL in multi-dimensional ways and to examine whether age, sex, education, health of childhood, BMI, residence, etc., are affecting the QoL directly or by multimorbidity indirectly. *Dolui et al.* (2023) revealed that in India rapidly increasing number of older adults, but life expectancy has been reduced, and non-communicable diseases like BP, diabetes, etc., are increasing. This results in many older adults suffering from multimorbidity, and this causes many physical and mental health diseases. The quality of life is defined by the WHO (1996) as “an individual’s perception of their position in life in the context of the culture and value systems in which they live and about their goals, expectations, standards and concerns”. And the research gap was studied. Multimorbidity indicates a poor quality of life, but the concept of multimorbidity, that how it connects the socioeconomic factors and QOL, was not clear. To overcome this gap, the study was done.

Data studied from the LASI wave-1 participants were 73,396, with 45+ years old. In this study, 14 chronic diseases were considered. To measure the QoL, it was divided into 6 domains: physical health, psychological health, social relationships, general health, life satisfaction, and environment. A total of 21 questions were included to measure the QOL. PCA (principal component analysis) is used to QoL score from 0 to 100. High score showed high QoL.

Descriptive and structural equation models (SEM) were conducted to find the prevalence and to understand the direct and indirect effects. Standardised and unstandardised regression coefficients (β) and p-values with 95% CI were used to determine significant direct, indirect, and total effects.

The equation for exogenous variables:

$$X_i = \lambda_{xi}\xi + \delta_i$$

Where X_i are the observed variables related to an exogenous latent variable. λ_{xi} are the factors loading for each observed variable on the latent construct. ξ represents the exogenous latent variable. δ_i represents the measurement error for each observed variable.

The equation for endogenous variables is:

$$Y_j = \lambda_{yj}\eta + \epsilon_j$$

Y_j were the observed variables associated with the endogenous latent variables. λ_{yj} are the factors loading for each observed variable on the endogenous construct. η represents the endogenous latent variables. ϵ_j represents the measurement errors for each observed variable.

SEM - equation of hypothesised causal relationships

$$\eta = B\eta + \Gamma\zeta + \zeta$$

Where η is the vector of the endogenous latent variables. B is a matrix of coefficients that represents the relationships among the endogenous latent variables. Γ is the matrix that represents the effect of exogenous variables on the endogenous variables. ζ represents exogenous latent variables. ζ is the vector of error terms affecting the endogenous variables.

Fit for the SEM model got checked through several measures: how X^2 compared to degrees of freedom (X^2/df), the GFI index, CFI, along with RMSEA. To pick the best structure, researchers used AIC - lower numbers here meant stronger alignment between model and data. All analyses were performed with R Studio version 2023.12.1 + 402, [1](#), SAS and STATA 16.

The results showed that 51% older adults had suffered from multimorbidity most common diseases being hypertension, gastrointestinal, and Musculoskeletal disorders. Multimorbidity was increasing with respect to the age factor, and QoL was reduced. Over the age of 75, the QoL was rapidly decreased. In women, multimorbidity was high, and QoL was low. Also, *Fortin et al.* (2007) and *Marengoni et al.* (2011), have consistently shown that multimorbidity significantly diminishes QoL across older populations. Daily physical activity was found to be less multimorbidity and a higher quality of life. Multimorbidity and QoL had a negative correlation. $\beta = -2.39$, i.e. Quality of life goes down as multimorbidity rises. What stands out is how other factors play through it. Older age, being male or female, body weight, money status, social group, and where someone lives - they do not affect quality of life directly. Instead, they feed into more chronic illnesses first. Those added conditions then pull life satisfaction lower. So the real pathway runs through multimorbidity. Therefore, only socioeconomic corrections were not sufficient to control multimorbidity in necessity.

This study concluded that multimorbidity majorly affects the quality of life in older adults. It increases the effect of socioeconomic and demographic factors. Therefore, integrated

geriatric healthcare, early screening, chronic disease management, and physical activity promotion were found to necessarily be updated.

1.3 Association of frailty with indoor air pollution among older adults and elderly population as per gender and age group: insights from Longitudinal Aging Study in India (LASI1st Wave)

Pritam Halder et al. (2025) wanted to find the problem of indoor air pollution and its effect on health. As the population increased day by day, frailty was found to be the major challenge in public health. Frailty is defined as a condition in which a reduction in physical fitness leads to an increase in the chances of multimorbidity, and leads to adverse outcomes such as falls, early mortality, disability, and hospitalisation. Halder M et al. (2024), found that 45+ older adults had a high rate of frailty; female, older adults, economically weaker, less educated, and urban people had a high chance of frailty. At the same time, India was suffering from the saviour condition of indoor air pollution, because in many houses, for cooking, coal, wood, and dung gas were used. However, Verma M et al. (2024) investigated the relationship between IAP and health conditions, but frailty was not measured in detail, and different age groups in females and males were not studied. To overcome these limitations, this study was conducted. This study mainly focused on showing the relationship between the IAP and frailty, and finding this in different age groups of females and males.

The data source was LASI Wave-1; this study included a total of 35 states and UTs in India, 45+ year old, and the final selection for analysis was 55,572 samples in size. It was based on a cross-sectional observational study. Because it shows association but does not verify the cause and effect.

The frailty was measured in five domains: continuous fatigue, Unintentional weight loss, weak grip strength, low physical activity, and slow walking speed. Participants consisting of 3 or more domains were detected as frail. IAP was not measured by only one question; however considered many questions related to multiple household indicators, including type of cooking fuel (clean vs. solid fuel) Type of stove (improved vs. traditional/open fire), Cooking location (inside or outside the house), Ventilation facilities (chimney, exhaust fan, window, or none), and Presence of household indoor smoking. These variables were taken together to construct the overall IAP exposure.

Results obtained approximately 17% of participants were known as the frail, and frailty was more likely detected in women, and 27% frailty was obtained in the 60+ age group, similar to other research findings Cao et al., (2022), Shin J et al.,(2021). In low education background, low income, rural, unemployed, and those without health insurance were commonly detected as frail. The association of IAP and frailty was noted as that solid fuels lead to a high risk of frailty, and poor ventilation was strongly associated with frailty. Total IAP exposure increased the 10-14% chance of frailty. According to gender, in males, solid fuels, poor ventilation, and indoor smoking were clearly associated with frailty. In females, only poor ventilation was found to be an important factor associated with frailty; other IAP indicators were low in numerically. With respect to age in the 45-59 age group, only poor ventilation was the main reason for frailty, and in the 60+ age group, all the indicators of IAP led to frailty.

The study indicates the importance of improving household environments through government initiatives such as PM Ujjwala Yojana (LPG), PMAY-G (home and ventilation), Ujala (LED), and tobacco control programs.

The study cannot clearly show cause and effect because it is based on people's self-reported answers. Some important factors, such as diet, genetic background, and place of residence, were not included in the analysis. Since participants had to recall past events, their answers may not always be accurate. However, the study included a large and diverse population from across the country, which strengthens the findings. Even so, dependence on memory may have influenced some of the results.

Older people in India, especially men, may become weaker due to breathing polluted air inside their homes. This study clearly shows that poor indoor air quality is an important but often ignored problem. Smoke from cooking and lack of proper ventilation can harm health over time. Using cleaner cooking fuels and improving airflow, such as opening windows, can reduce these risks. Public awareness and policies focused on indoor air quality can help older adults stay healthier. Although this issue is discussed less than diet or exercise, it is equally important, and small changes can lead to long-term benefits.

2.4 Association of perceived childhood socioeconomic status and health with depressive symptoms among middle-aged and older adults in India: using data from LASI Wave I, 2017–2018

Brahmanka et al. (2024). The study was conducted to determine whether there is any relation between childhood economic status and childhood health problems with depression in older adults. And to examine the society and health factors are the cause of depression or not. Because *Elder GH et al. (2018)* identified that depression is a serious problem, it reduces recall memory, quality of life, and mental health. According to the WHO and *Pilania M et al. (2016)* in 1 out of 6 is suffers from depression.

The data of the study were LASI Wave-1 (2017-2018), and the research design was a cross-sectional study. The sample size was 73,396, those aged 45+. The outcome variable was depressive symptoms, the CES-D scale, in which 10 questions were included using the previous studies examined by *Kumar S et al. (2016)*, and *Singh S et al. (2022)*, and the scores were given as 0 to 10. When 4 or more than 4, then they found that depression is present. Main causes were the predictor variables, which were given as childhood economic status, health of childhood, other controlling variables like age, gender, sex education, income, category, rural and urban, and multimorbidity.

Descriptive analysis was done to find the percentage, and logistic regression was used to look for the prevalence of the causal factor. The results were given in adjusted odds ratios (AOR).

Depression was found in 27.6% of 45+ aged, and age and depression were found to be directly proportional, because high depression was found in 80+ aged, women had higher depression than men, in economically weaker people had more depression than wealthy people. Those who were poor in childhood had a high chance of depression. And when someone suffered from a health problem in their childhood so there will be 1.56 times more chance to be depressed. High multimorbidity indicates a high risk of depression.

As the data was collected on information about childhood so it leads to recall bias, and a cross-sectional study was used, which shows some limitations of the study.

The study clearly observed that the status of economic and health in childhood highly affected depression and health in older age. Therefore, in childhood, nutrition should be better, better education and health facilities, so in future the mental problems can be reduced.

2.5 The burden of disease-specific multimorbidity among older adults in India and its states: evidence from LASI

Priyanka Patel et. al (2023) main objective of the study was to investigate the multimorbidity in various states in India. Older adults suffer from multiple chronic diseases, and it is

necessary to identify the socioeconomic factors related to multimorbidity. “As a result of higher lifespan and higher exposure to risk factors for chronic diseases, the burden of multimorbidity is quickly rising in India (Barik M et al. and Khan M et al., 2022)”.

They used the data from the LASI, and the total samples were taken, 31,464; 60+ older adults were included. In study 11, chronic diseases were considered, including hypertension, diabetes, cancer, lung diseases, heart diseases, stroke, arthritis, depression, cholesterol, thyroid and anaemia. The outcome variable was chronic morbidity. And the explanatory variables were introduced as sex, age, education, religion, marital status, wealth, region, and living arrangements. Descriptive analysis was done to find the prevalence and pattern of multimorbidity in older adults. The chi-square test was also used to find the significant relationship between socioeconomic variables and multimorbidity. Also, the multivariate analysis, a multinomial logistic regression (Kirchberger I et al., 2012), was carried out to find the factors that are associated with multimorbidity and its estimation was taken as the RR (risk relative ratio). The logit models group each response category with a baseline category; the most common model is

$$RR = \frac{P(Y = 1|X + 1)/P(Y = \text{base category}|X + 1)}{P(Y = 1|X)/P(Y = \text{base category}|X)}$$

This equation also indicates the effects of x on the dependent categories. And to calculate the probability of each category, the following equations were used.

$$P(1 - \text{one chronic disease}) = \frac{\exp(b_0 + b_1x_1(2) + b_nx_n(2))}{(1 + \exp(b_0 + b_1x_1(2) + b_nx_n(2)))}$$

$$P(2 - \text{multimorbidity}) = \frac{\exp(b_0 + b_1x_1(3) + b_nx_n(3))}{(1 + \exp(b_0 + b_1x_1(3) + b_nx_n(3)))}$$

Where P indicates the probability - b₀ was the constant, b₁ and b_n were the coefficients, and x₁ and x_n denote the number of explanatory variables.

The key findings were that the rate of multimorbidity in older adults in India is 32.1%. However, 30.3% people did not have a single chronic disease. Women had a higher chance of multimorbidity compared to men, and urban areas had a higher prevalence than rural areas. As age increased, the risk of morbidity also increased, especially in 75-79-year-old adults. On comparing the economic background, the wealthier older adults had a higher prevalence of multimorbidity. In southern India and western states, multimorbidity was found to be high

compared to other states, and Kerala was the highest among these states, with 59.2%. Those who had chronic heart disease in older adults had a second disease, mostly high 91% found.

The study concluded that with an aging population, the problem of multimorbidity becoming the most challenging health problem, particularly in heart chronic disease, stroke and thyroid patients were taken most care as the mostly affected by physical fitness. “There was substantial evidence on the regional variations in the prevalence of multimorbidity in India, attributed to differential healthcare access, utilisation, under-reporting and under-diagnosis (IIPS, NPHCE, HSPH, and USC, 2020)”

2.6 Prevalence, Factors and Inequalities in Chronic Disease Multimorbidity among Older Adults in India- Evidence from LASI

Chauhan al. (2022) studied the prevalence, patterns, associated factors, and socioeconomic inequalities of chronic disease multimorbidity among older adults in India. Patel R et al. (2019) showed that multimorbidity is a cause of serious challenges for healthcare systems because it reduces quality of life, and Araujo MEA et al. (2018) investigated higher disability and mortality among older adults with multimorbidity. Older people in India often suffer from more than one health problem at the same time, but health services usually treat only one disease at a time. Many studies have examined individual long-term illnesses, but there is limited information on multiple health problems occurring together and how they affect different social groups across the country. This gap motivated the present study. The aim of the study is to understand how common multiple long-term health conditions are among older Indians, identify the factors linked to them, examine which groups are most affected, and provide useful insights for developing more fair and effective health policies. From the first wave of the Longitudinal Aging Study in India - conducted between 2017 and 2018 - the information came. A total of 313,733 people aged sixty or above took part. This wasn't an experiment that followed changes over time; instead, it captured one moment across different lives. Each person shared details about their health and background through spoken responses. Nine long-term illnesses caught the researchers' attention: high blood pressure, issues with sugar regulation, problems affecting the heart, events involving blocked brain vessels, abnormal cell growth, breathing difficulties tied to lungs, joint pain conditions, mental or nerve-related challenges, plus elevated fat levels in the bloodstream.

Looking at how common multiple conditions were became clearer through basic comparison methods. To spot what links to having more than one health issue, they used a specific type

of math model that handles several outcomes at once. Also analysed the slope index of inequality and the relative index of inequality to measure the socio-economic inequalities.

They calculated that 24.1% of older adults had multimorbidity, and approximately 29% had a single chronic disease. The highly prevalent diseases were hypertension, arthritis and diabetes. Multimorbidity was higher among those aged 70 and above. Factors were found to be associated with a higher commonality of multimorbidity in gender women had a higher risk than men. In comparison, the rural and urban areas, urban show a higher prevalence of multimorbidity than rural areas. Related to education, a higher level of education was associated with a higher chance of multimorbidity. Wealthier older persons from wealthier households had higher odds of multimorbidity. And from the perspective of work status, those were unemployed had more chances of multimorbidity. These findings indicate that multimorbidity is not mainly a biological issue but is also highly affected by social and economic conditions. This study also carried out a strong association between multimorbidity and factors related to health. Older adults were more likely to have multimorbidity. People with ADL (activities of Daily living) and IADL (Instrumental activities of daily living) disabilities indicate much higher multimorbidity (**Roberts KC et al., 2015**).

The SII and RII measures, inequalities in multimorbidity, and prevalence of multimorbidity were more concentrated among women, urban residents, wealthier and more educated groups and individuals with disabilities.

The study has the strengths as it used a large sample from nationally representative data, it included multiple socio-economic and health variables and advanced inequality measures to include population-level disparities. And the limitations of the study were found chronic diseases were self-reported. It could be the cause of bias (**Ajaero CK et al, 2020**). Fewer than ten long-term illnesses made the list, yet the study's structure makes broad conclusions hard to justify. Though focused, its format holds back deeper insights.

Older people in India face growing challenges with multiple health conditions at once. Because money troubles and daily living difficulties often go hand in hand with these illnesses, care should shift from treating single diseases to supporting whole-person needs. What stands out is how social background shapes who gets sicker. Services must adapt - not just react - to keep up with aging populations.

2.7 Prevalence, Pattern, and Correlates of Multimorbidity among Adult and Old-Aged Women in India

Debsarma, Saha, and Choudhary (2022) focused on estimating the prevalence, disease patterns, and factors associated with multimorbidity among adult and older women in India. In India, women more commonly experience poor quality of health because of society's perspective, economic, and cultural phenomena, which increase their chance of getting chronic diseases (Fortin M, Haggerty, 2021). In previous studies in India, mostly conducted only on age groups or both genders together, instead of research mainly focusing specifically on women across adult and older ages, was still limited. This study objective to fill this gap.

The study was based on nationally level data from the second wave of the WHO Study on Global Aging and Adult Health (SAGE), introduced in 2015 in six Indian states, including Assam, Karnataka, Maharashtra, Rajasthan, Uttar Pradesh, and West Bengal. The final sample consisted used 4,898 women aged 18+ years. Multimorbidity was obtained based on self-conducted diagnosis with respect to 16 chronic diseases, given as hypertension, diabetes, arthritis, back pain, cataract, lung disease, heart disease, and mental health conditions. Statistical methods were carried out as descriptive analysis, chi-square tests, and multivariate logistic regression were done to study the prevalence and associated factors with multimorbidity.

The results indicate that multimorbidity was highly common among women in India. About **47.69%** of women had multimorbidity; however, **26.3%** had one morbidity. The commonality of multimorbidity was directly proportional to age. Women aged between 18 to 34 years had **14.6%** multimorbidity, but as age increased, the multimorbidity also increased to **34.8%** in women aged 35–49 years, and reached **65.6%** among women aged 70+ years. High prevalence rate found in the Back pain among both adult and older women, preceded by hypertension and arthritis. In older women aged 60+, cataract (30.2%), hypertension (27.3%), and arthritis (24.2%) were specifically common.

The study also found that the number of chronic diseases increased gradually with age. However, more than **57%** of women aged 18–34 years reported being healthy, and **12.8%** of women aged 70+ years were found to be disease-free. In 70+ years, **25.3%** of women aged 70 or older had four or more 4 chronic diseases. This clearly indicates that disease burden gradually collects over the lifetime. Also, (**2.1%**) of younger women had four or more chronic conditions, indicating that multimorbidity is not restricted to older age only. These findings corroborate with other studies, which indicated that biological aging influences the development of chronic diseases among women (**Hayek S et al. and Jaul E et al., 2017**)

Multivariate analysis obtained the results that age was the strongest factor of multimorbidity. This result was consistent in several other studies (**Alimohammadian M et al. and Puri P et al., 2017,2021**). Women aged 70+ years had **over 30 times higher chances** of having multimorbidity compared to women aged 18–34 years. Other important factors also identified were education, work status, perceived loneliness, and self-rated health. Women who suffer from loneliness have more than **twice times risk** of multimorbidity. Those who reported their health as poor were nearly **18 times more likely** to suffer from multimorbidity than healthy women. Working women also suffer from higher odds of multimorbidity due to work pressure and household responsibilities.

In conclusion, Debsarma et al. (2022) clearly concluded that multimorbidity is a major health problem among women in India, particularly among older age groups. The study indicates the need of the need for updated healthcare systems that control multiple chronic conditions together rather than focusing on a single morbidity.

2.8 Multimorbidity among Urban Poor in India: Findings from LASI, Wave-1

Sinha al. (2022) aimed to find the prevalence, patterns, associated factors, and results of multimorbidity in the urban poor in India. Information came from early results in a long-term Indian aging project. Having more than one ongoing health issue at once - called multimorbidity - shows up often here. This rise ties to quicker city growth, shifts in daily living, and older populations across nations such as India - not just distant reports but real patterns seen by researchers like (**Oni T et al., 2015**). The most unequal areas in India are urban areas, and there are several challenges faced by the poor populations living in cities with respect to accessing basic needs and healthcare services. This study was done to investigate the problem of multimorbidity in this group of population group and its implications for achieving universal health coverage.” The increasing burden of multimorbidity has a significant impact on the ability to achieve universal health coverage (UHC) (**Bukhman G et al. 2020**)”.

The study was conducted on nationally representative cross-sectional data collected during 2017–2018 from LASI Wave-1. The total of 9,327 of the poorer and poorest wealth groups in urban areas aged 45+ were considered. Multimorbidity was identified based on self-reported diagnosis of **17 chronic diseases**, consisting of hypertension, diabetes, heart disease, stroke, arthritis, chronic lung disease, oral morbidities, visual and hearing impairment, and

mental health conditions. Descriptive statistics and multivariable logistic regression were conducted to study prevalence and correlates.

The data revealed a high rate of multiple health conditions among city dwellers living in poverty - nearly half were affected. As people got older, these numbers climbed fast. Among adults between 45 and 59, about one in three had more than one illness. That jumped to slightly over half for those 60 to 74. People past 75 saw the sharpest rise, with nearly two out of three dealing with several chronic issues. Multimorbidity was high in females, individuals living single, those who were unemployed, and those from the poorest wealth group. Regional differences were also calculated, with the high commonality found in the eastern and southern regions of India.

In accordance with disease patterns, hypertension had the major role in multimorbidity. The most common disease combinations were **hypertension with other chronic diseases (3.44%)**, followed by **hypertension with diabetes (2.07%)**. These findings showed the importance of cardiometabolic and oral health conditions in introducing multimorbidity patterns among the urban poor.

The study also concluded that those who suffered from multimorbidity had higher (10.94%) chances of getting hospitalisation, **compared to 5.81% among those without** multimorbidity. Out-of-pocket expenditure showed a positive relationship with the number of diseases (**Bukhman G et al. and Garg CC et al., 2006, 2009**); the analysis showed that **INR 350 for those without any chronic disease to INR 1,080 for those with four or more conditions**. Those who had multimorbidity were considered as 10 times poorer quality of health than those who did not have a single disease.

In conclusion, Sinha et al. (2022) observed that multimorbidity is a crucial growing public health problem in the urban poor in India; therefore, those who had multimorbidity most commonly visited the hospitals, incurred the expenses on treatments, and had negative thoughts regarding their health condition. The study indicates the need for updated and equitable primary healthcare facilities, delivering the scheme like Ayushman Bharat to the urban poor groups.

2.9 Community prevalence and dyad disease pattern of multimorbidity in China and India - a systematic review”(BMJ Global Health, 2022).

Zhang et al. (2022) introduced a systematic approach to examine the community-level prevalence and disease patterns of multimorbidity in China and India. These days, folks are sticking around longer thanks to improved daily living conditions. Even so, living longer often means dealing with multiple long-term health issues together - what doctors call multimorbidity - which tends to wear down well-being, weaken strength, and sometimes bring about earlier passing. Therefore, the main focus of the study is on multimorbidity and to find its disease pattern. India and China are the top two countries in terms of population, according to **Pati S, Swain S, Hussain MA, et al (2023)**. **These countries together comprise about 36% of the world's population**, as investigated by **Wang LM and Mini GK (2019)** that both populations are aging rapidly. And multimorbidity is becoming common in these countries. **Violan C et al. (2014)** deduced clear and updated evidence about its prevalence, but which diseases occur together was missing. Previous studies conducted on the high-income countries, India and China such countries were less sysemantic reviewed on multimorbidity, and the information was not properly available. Looking into what has been missed so far, researchers wanted to see how widespread multiple health conditions are in adult populations across China and India. Instead of combining results statistically, they pulled together findings from different papers since methods varied too much. Their goal? Spotting frequent pairs of diseases among those dealing with more than one illness at a time. Patterns were checked against factors like age, gender, location, and national context. The work took shape as a structured overview - careful, organized, yet not aiming to pool numbers. How it ended up shaping understanding came down to consistency in gathering rather than merging data.

A search through Medline began the process, followed by results pulled from EMBASE. One after another, findings appeared in Cochrane reviews. Then came entries logged within Web of Science. All sources shared a common trait - they used English. Each played a role in spotting useful research. And the Chinese database: CNKI. Criteria included adults aged 18+, community-based studies (not hospital-based), conducted in China or India, patterns of multimorbidity, and published in English or Chinese. The initial total records studied or screened were 13,996; the final included studies were 59, in which 46 were taken from China, 9 from India, and from both countries were 4 studies. The quality of assessment was modified by the Newcastle-Ottawa Scale, and any assessment was found to be of high quality; most were in fair or poor quality.

The features of the study were that all **studies were cross-sectional**, sample sizes ranged from over 100 and nearly 2 million, and the number of diseases considered ranged from 2 to more than 40. The data were collected from primary surveys and secondary data (health records, national surveys).

The result of the studies shows that in the total population, multimorbidity was confirmed in 30.7% i.e., approximately one in three people suffer from multimorbidity. particularly in countries multimorbidity rate found in China was 36.1% and in India 28.3%. This clearly indicates that in compared to India, China was more commonly affected by multimorbidity. The result also verified that age and multimorbidity were positively correlated, and in the 60+ age multimorbidity was highest. Females had a higher prevalence than males because of the longer life expectancy of women, higher healthcare use by women, and inclusion of sex-specific diseases. And in urban and rural slight differences, but overall, similar prevalence. One common mix showed high blood pressure alongside diabetes, another linked it with joint pain, while some cases paired it with heart issues or metabolism problems - high blood pressure popped up nearly every time, making it stand out. Mood disorders like depression and anxiousness didn't show up as often in records, mostly because people weren't getting checked, feared judgment, or clinics skipped mental evaluations. In comparison to high-income countries, multimorbidity prevalence was found to be high in China, and India was lower **than in high-income countries**. The cause could be diagnosis, early death due to poor disease management, and fewer diseases included in studies.

This study has the limitation that there is high heterogeneity; therefore, a meta-analysis was not possible, and most of the studies were cross-sectional. There was a lack of standard introduction of multimorbidity, mental health, and HIV like conditions were not studied, and poor study quality overall.

This study clearly indicates that there must be a standard definition of multimorbidity, large-scale, representative studies, preventive strategies should be hypertension-centric, the middle-aged population is a high-risk group, and there must be better use of routine health data.

Overall, Zhang al. (2022) concluded that multimorbidity is highly prevalent in both China and India. The hypertension-based disease combinations were most common, although evidence was increasing, but there were methodological weaknesses and due to a lack of standardisation, variability in results. The future research should be high-quality, standardised, and policy-oriented.

2.10 Prevalence of multimorbidity and associated factors among older people in rural Northeast Brazil

Alexandrino et al. (2023) conducted a cross-sectional study to find the prevalence of multimorbidity and its associated factors in older adults living in rural northeast Brazil. In prospect of this study, multimorbidity was considered as when two or more chronic diseases present at the same time in an individual. Schneider J et al. (2021) and Ferretti F et al. (2019) studied that multimorbidity leads to many health challenges in lifestyle, such as an increase in mortality. This study was found to be important as multimorbidity was mostly found in older adults. Asante D et al. (2022) explored that most rural areas had insufficient medical facilities, hospitals were far away, and the level of education was low, with low income. But the research was limited based on rural older adults. Therefore, the two main objectives were for this study are to obtain the prevalence of multimorbidity in older adults and to identify the factors associated with social-behavioural and health-related to multimorbidity.

The research used a cross-sectional study, conducted at Aracagi (city), Paraiba (state) and the rural area of Brazil. An Initial 461 older adults were taken, used for analysis, 360 of the total 60+ years old. Multimorbidity was measured in binary form; if an individual had more than 2 diseases, then the status of multimorbidity was yes; if an individual had fewer than 2 chronic diseases, it shows multimorbidity was not present. Diseases were reported using the self-reported and a total of 32 chronic diseases were listed. Explanatory variables considered, such as social, behavioural, and health-associated factors.

Analysis was done using descriptive statistics showed the per cent, mean, and standard deviation. Chi-square test was done to find a significant relationship between multimorbidity and independent variables. Poisson regression was the main mathematical model of the study, considering multimorbidity as the outcome variable and a prevalence (>10% indicates a high prevalence rate). A simple equation was given as:

$$\log(\text{PR}) = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \dots + \beta_k X_k$$

Where **PR** = Prevalence Ratio, β_0 = Constant, and $X_1, X_2 \dots x_k$ are the explanatory variables. $\text{PR} = 1$ indicates that there is no relationship between outcome and independent variables, $\text{PR} > 1$, i.e., high risk or prevalence or multimorbidity, and $\text{PR} < 1$ shows less prevalence of multimorbidity.

From the study, they analysed that the prevalence of multimorbidity was 54.2% in older adults. Multimorbidity was found to be 16% higher in females than in males, aged the 70+ had a higher chance of multimorbidity than the 60-69 age group. Overweight older adults were found to have a higher commonality of multimorbidity than thin individuals, former smokers suffered from multimorbidity than non-smokers, and a higher chance of multimorbidity in those who did not drink alcohol than those who drank it. It was found to be reverse causality. Blood pressure issues showed up in just over half the cases - 60.3 percent to be exact. Diabetes came next, present in nearly one out of five people, sitting at 21.7 percent. Joints affected by arthritis appeared in 18.3 percent of those studied. A study by Maclean RR and team from 2017 points to tobacco and drinking problems as key factors you can actually change. These two together drive serious illness and death on a large scale. Every year, about 5 million lives end globally due to their combined impact.

Folks filled out surveys on their own, one snapshot in time, pulled together while the world dealt with lockdowns. Because of that setup, many could not join in. That gap shaped what the research could show.

Older folks living outside cities often deal with multiple health issues at once. Age, body weight, lifestyle choices, sex - these pieces tie closely to how medical help gets used. Picture a system built just for them, one that fits their real daily needs. Shifting focus toward staying well as people grow older could make a difference. Change might start by rethinking how care reaches remote places.

3.FORMULATION OF RESEARCH DESIGN

3.1 STATEMENT OF PROBLEM: Incidence of Multimorbidity and Its Related Factors Prevalent Among Aging Population in India: An in-depth Analysis Based on The Longitudinal Aging Study of India (LASI) Data.

3.2 AIMS AND OBJECTIVES:

- Descriptive analysis of the multimorbidity: Frequencies, cumulative frequencies, skewness and kurtosis, partition values, etc., reflects the distribution of multimorbidity in conjunction with several covariate factors, and its in-depth analysis leads to identification of the important covariate factors which have a significant impact on multimorbidity.

- Graphical representation of the data: The graphs will exhibit the prevalence trends and the distribution of multimorbidity by covariates like age, sex, education, and urban-rural setup, etc. The boxplots will also help in the detection of the outliers.
- Causal mediation: Assessment of how socio-economic factors mediate multimorbidity and how it impacts the functional human limitations or healthcare costs.
- Identification of multimorbidity clusters: Assessment of the multimorbidity patterns based on several subgroups of covariate factors.
- Development of MM index: Creation of a suitable weighted multimorbidity index that accounts for several biomarkers and self-reported health data (SRH).
- Survival analysis of the aging population of India based on multimorbidity and the related covariates.

3.4 Research questions

- What is the distribution and overall pattern of multimorbidity among older adults in India?
(Using descriptive measures such as frequencies, skewness, kurtosis, and partition values.)
- How does multimorbidity vary across key socio-demographic factors such as age, sex, education, and urban–rural residence?
- What role do socio-economic factors play in influencing multimorbidity and its related outcomes, such as functional limitations and healthcare costs?
- Do socio-economic factors mediate the relationship between multimorbidity and functional health outcomes among older adults?
- Are there distinct clusters or patterns of multimorbidity among older adults based on their health conditions and background characteristics?
- How does multimorbidity affect survival among the aging population in India, after accounting for relevant covariates?

.3.4 DESCRIPTION OF DATA COLLECTION:

This study is based on secondary data taken from an open-access source called the Longitudinal Ageing Study of India (LASI). The data were collected by the International Institute for Population Sciences (IIPS) during 2017–18. LASI provides detailed information about people's health, economic conditions, and social background. More details about the dataset are explained in Subsection 1.2.

4. RESEARCH METHODOLOGY:

4.1 Statistical tools and techniques:

1. **Frequency Distribution:** Frequency distribution is a representation of the values to show how many times the observation occurred in a particular category or class interval. It is often represented by a table or graphic using a histogram, bar, or polygon. It is used because when observations, discrete or continuous, are available on a single characteristic of a large number of individuals, it often becomes necessary to condense the data as far as possible without losing any information of interest.
2. **Measures of central tendency:** They provide an idea about the concentration of the values in the central part of the distribution.
 - 2.1 **Mean :** Mean of a set of values is their sum divided by the number of observations. The mean provides a single representative value of the dataset and is useful for comparing overall levels of variables across groups.
 - 2.2 **Median:** Median of the distribution is the value of the variable which divides it into two equal parts. It is the value such that the number of observations above it is equal to the number of observations below it. Thus, the median is a positional average or half-quartile.
 - 2.3 **Mode:** Mode is the value which occurs most frequently in a set of data and around which the other items of the set cluster densely. It is especially useful for categorical or discrete data.
3. **Skewness:** Skewness measures the degree of asymmetry in the distribution of a variable around its mean.
A distribution is positively skewed when the tail extends to the right and negatively skewed when the tail extends to the left.

We used skewness to understand whether variables such as age, income, or disease count were symmetrically distributed or showed concentration toward lower or higher values.

4. Kurtosis: Kurtosis describes the **peakedness or flatness** of a data distribution relative to a normal distribution.

A distribution with **high kurtosis (leptokurtic)** indicates heavy tails and a sharp peak, while **low kurtosis (platykurtic)** reflects lighter tails and a flatter distribution.

We examined kurtosis to assess the presence of extreme values and tail behavior.

5. Quartiles: Quartiles divide the dataset into **four equal parts**: the first quartile (Q1), median or second quartile (Q2), and third quartile (Q3).

They provide summary of the **central tendency and dispersion** of the data.

We used quartiles to describe the spread of key variables such as income, age, and disease count, and to support graphical tools like boxplots for identifying variability and outliers in the study.

6.Boxplot (Box-and-Whisker Plot)

A boxplot is a graphical display that summarises data using the minimum, first quartile (Q1), median (Q2), third quartile (Q3), and maximum.

We used it to detect outliers in the data. It is especially useful for comparing distributions on the basis of age groups and sex.

7. Bar Chart

A bar chart represents categorical data using rectangular bars with heights proportional to frequencies or values.

We used it to compare quantities across different categories clearly and visually, for example, to compare multimorbidity between males and females.

8. Stacked Bar chart:

A stacked bar chart shows bars divided into sub-categories stacked on top of each other, representing parts of a whole. We used it to see the contribution of each subgroup (say, males and females) within a category.

9. Histogram

A pie chart is a circular chart divided into slices, where each slice represents a proportion of the total. It is used to show the percentage or proportional distribution of categories when the number of categories is small.

10. Pie Chart:

A pie chart is a circular chart divided into slices, where each slice represents a proportion of the total. It is used to show the percentage or proportional distribution of categories when the number of categories is small.

11. Logistic Regression:

Logistic Regression is a statistical method that is used to model the relationship between one or more independent variables and a binary (dependent) outcome.

Logistic Regression models the probability that the dependent variable takes a particular outcome (0/1) as a function of explanatory variables.

12 Use of McFadden R^2 :

It is the coefficient of determination that is used to assess the goodness of fit of models in the case of logistic regression. Unlike the usual R^2 in linear regression, which measures the proportion of variation explained by the model, McFadden R^2 is based on a comparison of the log-likelihoods of the fitted model and the null model (model with no predictors). Hence, mathematically it is defined as;

$$R_F^2 = 1 - \frac{\log L_{model}}{\log L_{null}}$$

13. Odds Ratio (OR):

The odds ratio is the ratio of the odds of the occurrence of an event under one condition to the odds of its occurrence in another condition. It measures the strength and the direction of association between a particular predictor and the outcome.

- If OR takes the value 1 (OR=1), it signifies that there is no association between the predictor and the outcome; the odds remain unchanged.
- An odds ratio greater than 1 (OR>1) indicates that the odds of the event increase as the predictor increases, signifying a positive association.
- An odds ratio less than 1 (OR<1) indicates that the odds of the event decrease as the predictor increases, signifying a negative association.

14. Confidence Interval of OR (correct)

A 95% confidence interval for the odds ratio is given by:

$$(OR \cdot e^{-1.96SE}, OR \cdot e^{1.96SE})$$

Where $SE(\ln(OR)) = \sqrt{\frac{1}{a} + \frac{1}{b} + \frac{1}{c} + \frac{1}{d}}$ where, In a 2×2 table: a = exposed with outcome, b = exposed without outcome, c = unexposed with outcome, and d = unexposed without outcome.

15. Factor Analysis:

Factor analysis is a multivariate dimension reduction statistical technique used to examine the underlying structure of a large set of observed variables by identifying a smaller number of unobserved (hidden) variables, known as latent factors. It assumes that the correlations among the observed variables occur because they are influenced by these common latent factors. Loadings are the coefficients that show how strongly each original variable contributes to a factor.

16. Tetrachoric correlation matrix:

A tetrachoric correlation matrix is used when all the observed variables are binary (0/1). Instead of using ordinary Pearson correlations, which are not appropriate for binary data. Tetrachoric correlation assumes that these binary variables arise from underlying continuous variables. Hence, tetrachoric correlations estimate what the correlation would be if the underlying continuous variables were observed.

17. Parallel Analysis:

Parallel analysis works by creating many random datasets that have no real structure but have the same number of variables and observations as the original data (simulated data). Eigenvalues are hence calculated using this simulated dataset to represent what would be expected purely by chance, which are further compared with the eigenvalues from the actual data. If an eigenvalue from the real data is larger than the corresponding random eigenvalue, it suggests that the factor captures meaningful information; otherwise, that variable is considered insignificant.

18. Scree Plot:

A scree plot is a graphical method used in factor analysis to determine the appropriate number of factors or components to retain, which represent the maximum amount of variance explained. It is constructed by plotting the eigenvalues in descending order against the corresponding factor or component numbers. Typically, the point at which there is a sharp decline in slope is often referred to as the elbow, which indicates the number of significant factors to keep, while the remaining components contribute relatively little to explaining the variation in the data and are usually considered insignificant.

4.2 DESCRIPTION OF VARIABLES:

The following variables were taken directly from the LASI data:

VARIABLES COVERED IN THE LASI DATA	
Variable Name	Description
age	Age of the respondent in completed years
sex	Gender of the respondent
marital_status	Current marital status of the respondent
education	Highest level of education completed
employment_status	Current employment or work status
household_size	Total number of members in the household
self_rated_health	Respondent's self-assessment of their health
chronic_conditions	Presence of diagnosed chronic diseases
adl_difficulty	Difficulty in activities of daily living
iadl_difficulty	Difficulty in instrumental activities of daily living
mental_health	Indicators of depression and psychological well-being
cognition_score	Composite score measuring cognitive functioning
health_insurance	Whether any health insurance covers the respondent
healthcare_utilization	Use of healthcare services in the recent period
smoking_status	Current and past smoking behaviour
alcohol_consumption	Frequency of alcohol use
physical_activity	Engagement in physical or exercise activities in their daily lives
income	Individual or household income
wealth_index	Composite index of household assets and wealth
pension	Receipt of pension or old-age benefits
social_participation	Participation in social or community activities
living_arrangement	Household living arrangement of the respondent
caregiving	Provision or receipt of caregiving support by family or professionals
migration_history	Migration status and history
religion	Religious affiliation
caste	Caste or social group classification
residence	Urban or rural residence
state	State of residence

TABLE 4.2.1 Variables included in the study from the

Variable Name	Transformed Variable	Indicator/Categories/ Description
Age	i_age	We have categorized the age in four categories: <40 years, 40-59 years, 60-79 years and ≥ 80 years.
Sex	i_sex	Gender is transformed in dichotomous variable i.e 1 (Male) and 0 (Female).
Marital Status	i_ms	It is transformed into a trichotomous variable as 0(currently married), 1(widowed) and 2(divorced/separated/never married)
Education	i_edu	It is indicated as 0(no education), 1(primary), 2(secondary) and 3(higher).
Employment Status	i_emp	these values indicates as 0(never worked), 1(not working currently) and 2(currently working).
Self-Rated Health	i_srh	It is transformed into a trichotomous variable as 0 (Good (excellent/very good/good)), 1(fair) and 2(poor).
Health Insurance	i_health_ins	It is transformed into a dichotomous variable as 0(have insurance) and 1(does not have insurance).
Smoking Status	i_smoke	It is transformed into a dichotomous variable as 0(does not smoke) and 1(smokes).
Alcohol Consumption	i_alcohol	It is transformed into a dichotomous variable as 0(does not consumes alcohol) and 1(consumes alcohol).
Physical Activity	i_physical	It is indicated as 0(low), 1(moderate) and 2(high).
Pension	i_pension	0 indicates not receiving pension and 1 indicates receiving pension.
Residence	i_reside	0 indicates rural and 1 indicates urban.
Count of diseases	mm	It indicates the total number of chronic diseases. It ranges from 0 to 6 where if the count of diseases ≥ 6 then also mm=6.
Multimorbidity	i_mm	0 indicates no multimorbidity and 1 indicates presence of multimorbidity.

TABLE 4.2.2 Definition of Transformed Variables

However, for our analysis, we need to transform the data suitably and define the following new variables;

The disease count variable (**mm**) was constructed by summing the presence (1 = yes, 0 = no) of the following conditions:

- Jaundice / Hepatitis
- Tuberculosis
- Malaria
- Diarrhoea
- Typhoid
- Urinary tract infection (UTI)
- Anaemia

- Chikungunya
- Dengue
- Other infectious diseases
- Immunity-related conditions
- Angina
- Childhood health conditions
- Uterine conditions
- Use of supportive devices
- Hearing impairment / hearing aid
- High blood pressure
- Diabetes
- Cancer
- Chronic lung disease
- Heart disease
- Stroke
- Arthritis
- High cholesterol

Since the majority of the elderly in the sample. In the case of gender distribution, 1 represents male and 0 represents female. Since the elderly in the sample lie between 50 to 70 years therefore we censor the observation for the elderly having age >70 years. Annual household income was categorised into six groups based on the responses:

Category	Income Range (₹)	Income Group
1	0 – 100,000	Low income
2	100,001 – 300,000	Lower-middle income
3	300,001 – 600,000	Middle income
4	600,001 – 1,000,000	Upper-middle income
5	1,000,001 – 2,500,000	High income
6	> 2,500,000	Very high income

TABLE 4.2.3 Income Distribution

Since the majority of elderly sample range between 47 to 70 years, therefore we censored the observation for the elderly for >70

4.3 ANALYSIS:

We carried out the analysis on the LASI data using the independent variables obtained in TABLE 4.2.2. In this subsection, we carry out a descriptive analysis and observe the

association between various independent variables and MM to gain deeper insights into our data, moving forward with regression analysis and factor analysis and lastly to compute the MM Index.

4.3.1 DESCRIPTIVE ANALYSIS

4.3.1.1 Analysis of family size.

the detailed description of family size variable is given from Table 4.3.1.1, and the figure 4.3.1.1 bar chart. All measures of central tendency are given with the variability of the variable or dispersion as follows. Total samples are considered 73396 mean is found to be 3.0423. This indicates the number of family members commonly spreads around 3 size of a family. Median is 3, i.e. households size 3 divides the data of family size into two equal parts. The most frequent family size is as mode appeared 3. Variance is 0.611, which indicates that in family size members variability is not high. A bit off center at 0.041, the skew shows at symmetry. With kurtosis obtained below 3 - specifically 0.952 - the peak stays modest, neither too sharp nor flat. Spans from just one person up to five define family size here. At the 25th, 50th, and 75th marks, each lands exactly on three.

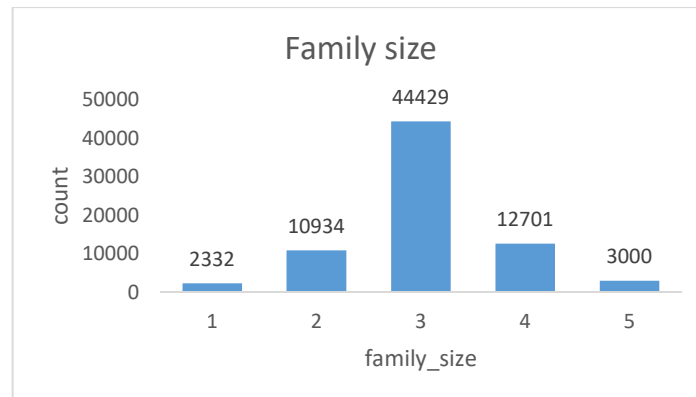


Figure 4.3.1.1.1 frequency distribution of family size

4.31 Family Size Statistics

family_size

N	Valid	73396
	Missing	0
Mean		3.0423
Median		3.0000
Mode		3.00
Std. Deviation		.78156
Variance		.611
Skewness		.041
Std. Error of Skewness		.009
Kurtosis		.952
Std. Error of Kurtosis		.018
Range		4.00
Minimum		1.00
Maximum		5.00
Sum		223291.00
Percentiles	25	3.0000
	50	3.0000
	75	3.0000

Table 4.3.1.1.1 descriptive analysis of family size

Now comparing it with multimorbidity, from Figure 4.3.1.1, it can be verified that the frequency of multimorbidity is high in the family size of 3 members, and lowest in the size of 1 member. This indicates that family size has a limited but noticeable association with multimorbidity, where larger families tend to have slightly more multimorbidity cases. And the fig 4.3.1.2 pie chart shows that multimorbidity is fairly evenly spread across different family sizes. Family size 4 has the highest share (23%). Family sizes 1 and 2 each account for 20%. Family size 5 represents 19%, while family size 3 has the lowest share (18%). Overall, there is no big difference in multimorbidity across family sizes.

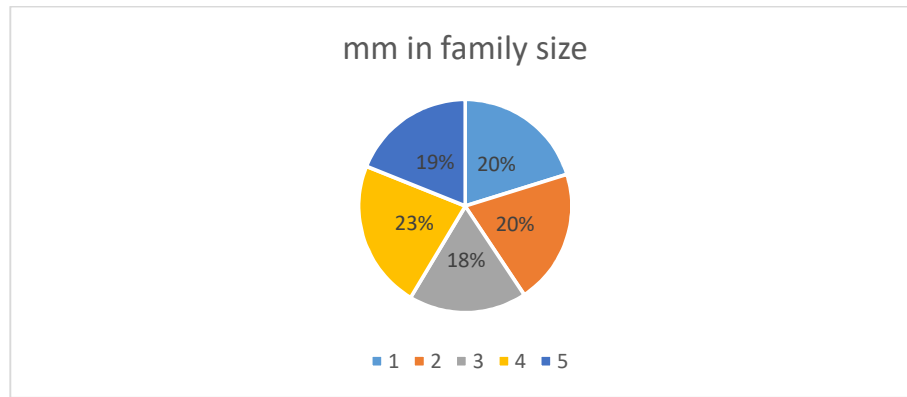


Fig 4.3.1.1.2 percentage of mm wrt to family size

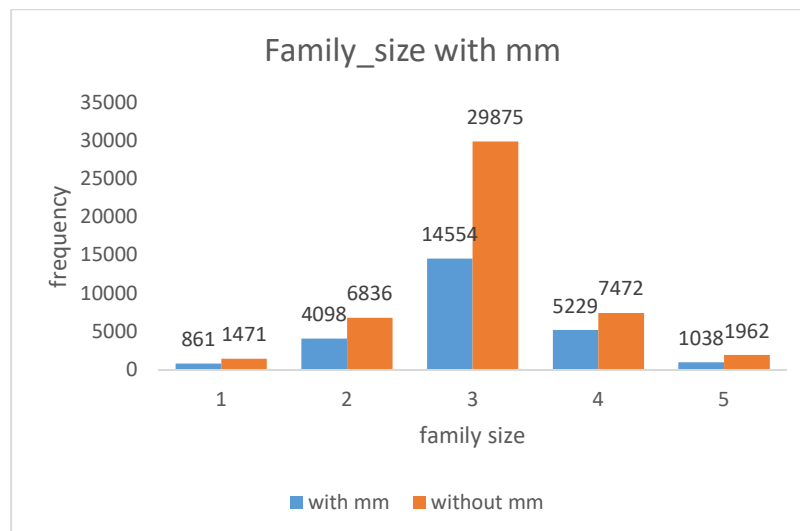


Fig 4.3.1.1.3 comparison of family size with MM

4.3.1.2 Analysis of rural and urban.

Frequency distribution of rural–urban residents is shown in Figures in 4.3.1.2.1 around 60% elderly population is rural residents. Fig 4.3.1.2.2 bar chart indicates that, in absolute terms, the number of individuals with multimorbidity is higher in rural areas than in urban areas, which is likely attributable to the larger rural population in the sample. A similar pattern is observed among individuals without multimorbidity, with rural areas reporting higher frequencies than urban areas. In contrast, Fig 4.3.1.2.3 pie chart presents the proportional distribution of multimorbidity and reveals that 52% of multimorbidity cases are concentrated in urban areas, while 48% are in rural areas.

Taken together, these findings suggest that although rural areas contribute a greater absolute number of multimorbidity cases, the relative burden of multimorbidity is marginally higher in urban areas. This highlights the importance of considering both absolute counts and proportional measures when assessing urban–rural differences in multimorbidity prevalence.

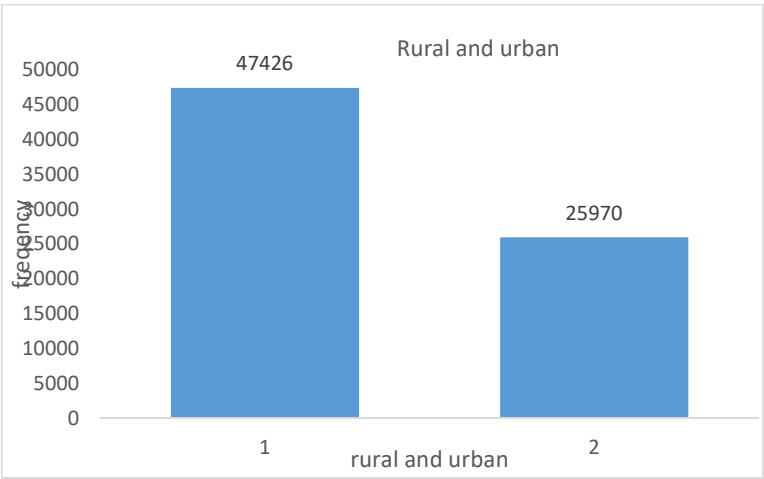


Fig 4.3.1.2.1 count of rural and urban

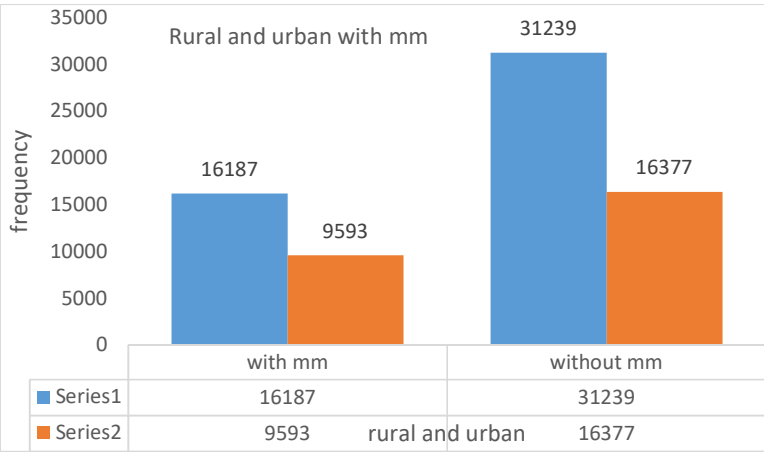


Fig 4.3.1.2.2 rural and urban frequency with mm

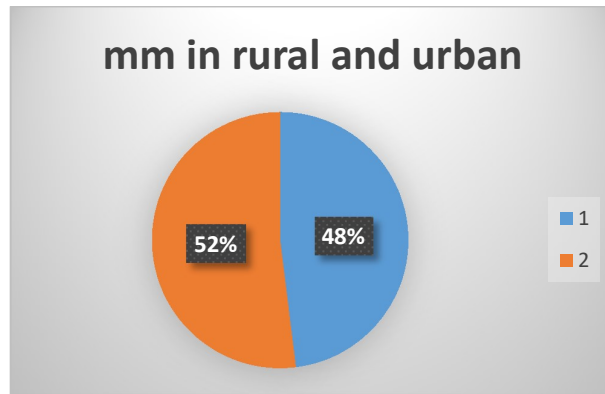


Fig 4.3.1.2.3 comparison of mm in rural and urban

4.3.1.3 Analysis of education levels with MM

Table 4.3.1.2 presents the education levels of 73,396 individuals, with no missing information. The mean education score is 0.94, which is close to primary education, indicating that most individuals have only basic schooling. The median value of 1 shows that half of the population has primary education or less, while the mode of 0 reveals that no education is the most common category. The standard deviation of 1.027 suggests noticeable variation in education levels, ranging from no education (0) to higher education (3). The positive skewness value (0.691) indicates that the distribution is slightly right-skewed, meaning a larger proportion of individuals are concentrated at lower education levels, with fewer attaining higher education. The negative kurtosis (-0.804) shows a relatively flat distribution. Percentile values further support this pattern, as at least 25% of individuals have no education, 50% have primary education or less, and only about 10% have reached higher education.

education

N	Valid	73396
	Missing	0
Mean		.94
Median		1.00
Mode		0
Std. Deviation		1.027
Variance		1.055
Skewness		.691
Std. Error of Skewness		.009
Kurtosis		-.804
Std. Error of Kurtosis		.018
Range		3
Minimum		0
Maximum		3
Sum		68683
Percentiles	25	.00
	50	1.00
	75	2.00

Table 4.3.1.3.1 description of education level.

From the table 4.3.1.3.1 and table 4.3.1.3.2 we can clearly observe that Nearly 46% of the people have no education, which is the largest group. **About** 24.6% have primary education, **while** 19.3% have secondary-level education. Only 10.2% of individuals have higher education, making it the smallest group. The cumulative percentages indicate that around 71% of the population has primary education or less, and nearly 90% have not gone beyond secondary education. Overall, the table clearly shows that most people belong to lower education levels, and relatively few have higher education.

education

	Frequency	Percent	Valid Percent	Cumulative Percent
Valid 0	33762	46.0	46.0	46.0
1	18037	24.6	24.6	70.6
2	14145	19.3	19.3	89.8
3	7452	10.2	10.2	100.0
Total	73396	100.0	100.0	

Table 4.3.1.3.2 Frequency of education level

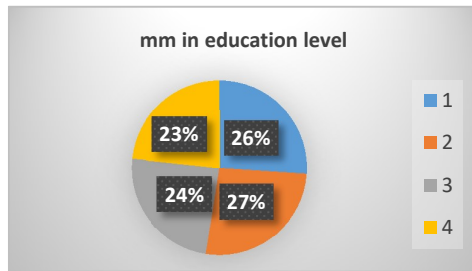


Fig4.3.1.3.1 pie chart mm in education level

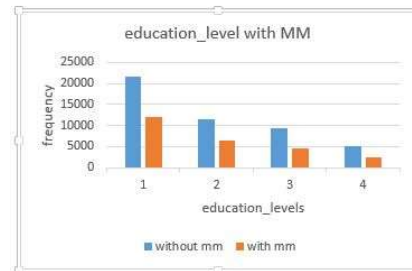


Fig 4.3.1.3.2 bar chart of mm in edu_level

4.3.1.4 TESTING THE ASSOCIATION BETWEEN THE INDEPENDENT VARIABLES AND MULTIMORBIDITY

a. To test the association between multimorbidity and residence(rural=0, urban=1), we applied a chi-square test followed as following. The test hypothesis is as follows:

H0: There is no significant relationship between residence and multimorbidity.

H1: There is a significant relationship between residence and multimorbidity.

Chi-Square Tests					
	Value	df	Asymptotic Significance (2-sided)	Exact Sig. (2-sided)	Exact Sig. (1-sided)
Pearson Chi-Square	58.054 ^a	1	.000		
Continuity Correction ^b	57.930	1	.000		
Likelihood Ratio	57.861	1	.000		
Fisher's Exact Test				.000	.000
Linear-by-Linear Association	58.053	1	.000		
N of Valid Cases	73396				

a. 0 cells (0.0%) have expected count less than 5. The minimum expected count is 9121.84.

b. Computed only for a 2x2 table

Table 4.3.1.4.1 chi-square test between mm and residence

From the above we can observe that a significant relationship exists between multimorbidity and rural–urban residence since ($\chi^2(1) = 58.054$ and $p=0 < .05$). This suggests that multimorbidity distribution varies significantly between rural and urban population.

b. To test the association between multimorbidity and Family size, the test hypothesis is as follows:

H0: There is no significant relationship between Family size and multimorbidity.

H1: There is a significant relationship between Family size and multimorbidity.

Chi-Square Tests			
	Value	df	Asymptotic Significance (2-sided)
Pearson Chi-Square	343.186 ^a	4	.000
Likelihood Ratio	339.435	4	.000
Linear-by-Linear Association	15.279	1	.000
N of Valid Cases	73396		

a. 0 cells (0.0%) have expected count less than 5. The minimum expected count is 819.10.

Table 4.3.1.4.2 chi-square test between mm and Familysize

From the above table we can conclude that there is a highly significant relationship ($\chi^2(4) = 343.186$, $p < .05$). The prevalence of multimorbidity varies across different family sizes, indicating it is not evenly distributed.

4.3.2 REGRESSION ANALYSIS:

To find the significant variables of multimorbidity, a binary logistic regression model was used. Here, the dependent variable (i_mm) is dichotomous, 1 denotes the presence, and 0 denotes the absence of multimorbidity, and the model estimates the log-odds of multimorbidity as a function of our independent covariates such as age, gender, occupation, residence, exercise, drinking, and self-reported health.

The fitted model is expressed as:

$$\log\left(\frac{p}{1-p}\right) = \beta_0 + \beta_1 I_{age} + \beta_2 I_{gender} + \beta_3 I_{occupation} + \beta_4 I_{residence} \\ + \beta_5 I_{exercise} + \beta_6 I_{alcohol} + \beta_7 I_{srh} + \beta_8 I_{edu} + \beta_9 I_{pension} \\ + \beta_{10} I_{family}$$

Where p represents the probability of having multimorbidity and β denotes the corresponding coefficients.

In this model, we have selected various socio-demographic, household, and lifestyle factors as our independent variables, which are also dichotomous. These independent variables are defined as:

I_age (Age): It takes value 1 when age ≥ 65 and 0 otherwise;

I_edu (Education); It takes value 1 if an individual has completed at least middle school and 0 otherwise;

I_occupation (Occupation); It takes value 1 if individual is in workforce and 0 otherwise;

I_residence (Place of residence); It takes value 1 if an individual reside in urban area and 0 otherwise;

I_gender (Gender); It takes value 1 for males and 0 for females;

I_srh (Self reported health); It takes value 1 if score ≥ 3 and 0 otherwise;

I_alcohol (Alcohol consumption); It takes value 1 if a person consumes alcohol and 0 otherwise;

I_exercise (Exercise); It takes value 1 if an individual exercises for at least 30 minutes per day and 0 otherwise;

I_pension (Pension); It takes value 1 if an individual receives pension and 0 otherwise;

I_family (Family size); It takes value 1 if there are at least 2 members in the family and 0 otherwise.

Based on the output obtained by running logistic regression using R we obtain the following results:

- The McFadden $R^2 = 0.0485$, which indicates that the model explains a modest proportion of variation in multimorbidity.
- The estimates of the parameters of logistic regression are interpreted in terms of the odds ratio (OR).

INDEPENDENT VARIABLES	ODDS RATIOS	INTERPRETATION OF THE ESTIMATED COEFFICIENTS
Age	1.388	The odds of an individual being multimorbid increases 2% for every one-year increase in age while keeping the remaining independent variables fixed.
Gender	0.914	Males have about 8.6% lower odds of multimorbidity compared to females, controlling for other independent variables.
Education	0.956	The odds of being multimorbid for educated individuals are 0.956 times the odds for non-educated individuals.
Family Size	0.992	The odds of being multimorbid for individuals in families with ≥ 2 members are 0.992 times the odds for those in smaller families.
Occupation	0.771	Working individuals have about 22.9% lower odds of being multimorbid compared to non-working individuals
Urban Residence	1.119	Urban individuals have about 11.9% higher odds of being multimorbid compared to rural individuals.
Self- Reported Health	0.32	Individuals with good self-reported health have about 68% lower odds of multimorbidity compared to those with poor health.
Exercise	1.008	The odds of being multimorbid for people who exercise are 1.008 times the odds for those who do not.
Alcohol consumption	1.068	Individuals who consume alcohol have about 7% higher odds of multimorbidity.
Pension Receipt	1.191	Individuals receiving pension have about 19% higher odds of multimorbidity.

TABLE 4.3.2.1 Computation of Odds Ratio and its Interpretation

4.3.3 TESTS FOR SIGNIFICANCE OF ODDS RATIO:

In order to test the null hypothesis $H_0: OR=1$ we have constructed 95% confidence intervals for the odds ratios in the following table to test for the significance of the OR w.r.t each of the independent variables:

INDEPENDENT VARIABLES	CONFIDENCE INTERVAL	INTERPRETATION
Age	(1.339, 1.439)	Reject H_0 ; $OR > 1$
Gender	(0.879, 0.950)	Reject H_0 ; $OR < 1$
Education	(0.924, 0.990)	Reject H_0 ; $OR < 1$
Family Size	(0.972, 1.012)	Accept H_0 ; $OR = 1$
Occupation	(0.744, 0.800)	Reject H_0 ; $OR < 1$
Urban Residence	(1.082, 1.158)	Reject H_0 ; $OR > 1$
Self-Reported Health	(0.307, 0.334)	Reject H_0 ; $OR < 1$
Exercise	(1.007, 1.009)	Reject H_0 ; $OR > 1$
Alcohol consumption	(1.020, 1.120)	Reject H_0 ; $OR > 1$
Pension Receipt	(1.099, 1.290)	Reject H_0 ; $OR > 1$

TABLE 4.3.3.1 Testing of significance of OR

4.3.4 FACTOR ANALYSIS:

To understand the patterns of Multimorbidity, the factor analysis was performed using the tetrachoric correlation matrix, which is appropriate for binary health variables, viz., BP, diabetes, cancer, lung disease, heart disease, stroke, arthritis, high cholesterol, and angina.

The following matrix was obtained using R:

Variable	High BP	Diabetes	Cancer	Lung Disease	Heart Disease	Stroke	Arthritis	High Cholesterol	Angina
High BP	1	0.5	0.11	0.16	0.41	0.38	0.23	0.46	0.19
Diabetes	0.5	1	0.12	0.09	0.31	0.25	0.14	0.43	0.06
Cancer	0.11	0.12	1	0.05	0.15	0.12	0.05	0.2	0.09
Lung Disease	0.16	0.09	0.05	1	0.23	0.09	0.19	0.15	0.28
Heart Disease	0.41	0.31	0.15	0.23	1	0.28	0.17	0.43	0.45
Stroke	0.38	0.25	0.12	0.09	0.28	1	0.12	0.23	0.1
Arthritis	0.23	0.14	0.05	0.19	0.17	0.12	1	0.21	0.21
High Cholesterol	0.46	0.43	0.2	0.15	0.43	0.23	0.21	1	0.13
Angina	0.19	0.06	0.09	0.28	0.45	0.1	0.21	0.13	1

TABLE 4.3.4.1 Tetrachoric Correlation Matrix

This correlation matrix indicates a moderate association among diseases:

- high BP with diabetes and high cholesterol,

- heart disease with angina and
- high cholesterol and stroke with high BP.

This suggests that there is a presence of cluster comorbid conditions, particularly associated with cardiovascular health. We conducted a parallel analysis to find the number of factors.

The following two-factor solution was obtained using the principal axis factoring with oblimin rotation, allowing factors to be correlated ($cutoff = 0.3$).

Variable	PA1	PA2
high bp (0 = no)	0.744	—
diabetes (0 = no)	0.687	—
cancer (0 = no)	—	—
lung disease	—	0.333
heart disease	0.446	0.409
stroke	0.425	—
arthritis	—	—
high cholesterol	0.644	—
angina	—	0.817
Measure	PA1	PA2
Proportion of Variance	0.212	0.112

TABLE 4.3.4.2 Output of Factor Analysis

The following scree plot obtained using R validates our result obtained from parallel analysis.

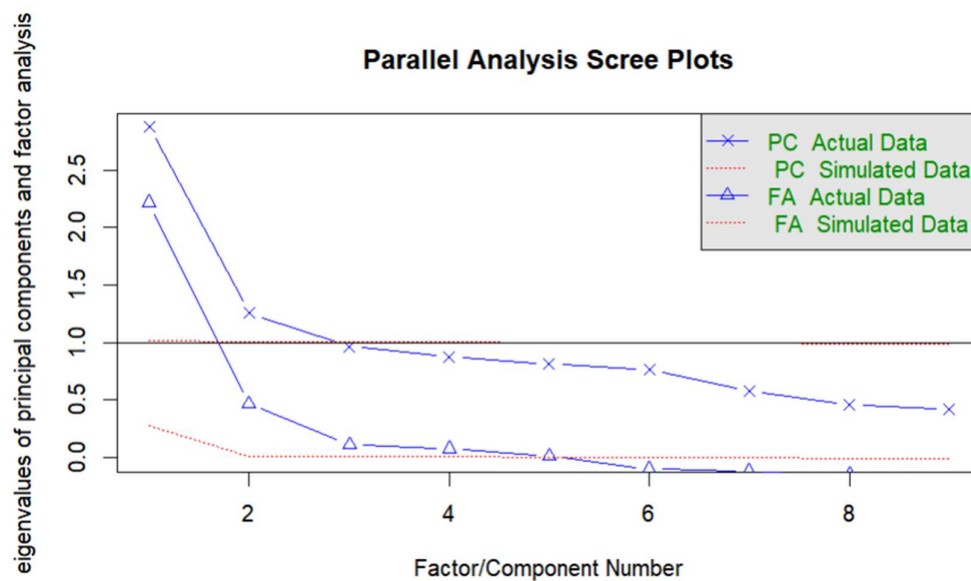


FIG 4.3.4.1 Scree Plot

Prominently, two factors emerge from the analysis, namely:

- FACTOR-1, which represents **cardiometabolic risk cluster** including diseases: High blood pressure, Diabetes, High cholesterol, Heart disease and Stroke. This factor includes those diseases that commonly co-occur due to risk factors such as ageing, lifestyle, and metabolic dysfunction.
- FACTOR-2, which represents **cardiorespiratory risk cluster** including diseases Angina, Heart disease and Lung disease. This factor links heart-related diseases with respiratory diseases.

Factor 1 explains 21.2% of the variation, Factor 2 explains 11.2% of the variation, and together, 32.5% of the total variance is explained by these factors.

To further understand how these factors are influenced by various independent variables, factor analysis was conducted using the data on the diseases corresponding to those 9 variables and where i_mm takes the value 1. By applying the filter, $i_mm=1$ and $I_gender=1$ as well as $i_mm=1$ and $I_gender=0$, the factor loadings obtained are as given below:

Variable	PA1	PA2	Variable	PA1	PA2
high.bp (1 = no)	0.642	—	high.bp (1 = no)	0.541	—
diabetes (1 = no)	0.642	—	diabetes (1 = no)	0.599	—
cancer (1 = no)	—	—	cancer (1 = no)	—	—
lung disease	—	—	lung disease	—	—
heart disease	—	0.99	heart disease	—	0.899
stroke	—	—	stroke	—	—
arthritis	—	—	arthritis	—	—
high cholesterol	0.369	0.309	high cholesterol	0.397	—
angina	-0.551	—	angina	-0.52	0.312
Measure	PA1	PA2	Measure	PA1	PA2
Proportion of Variance	0.158	0.133	Proportion of Variance	0.129	0.113

TABLE 4.3.4.3 Factor Loadings for Males (left) and Females (right)

Similarly, by applying two filters viz. ($i_mm=1$ & $I_age=1$) and ($i_mm=1$ & $I_age=0$) we get,

Variable	PA1	PA2	Variable	PA1	PA2
high.bp (1 = no)	0.61	—	high.bp (1 = no)	0.561	—
diabetes (1 = no)	0.607	—	diabetes (1 = no)	0.623	—
cancer (1 = no)	—	—	cancer (1 = no)	—	—
lung disease	—	—	lung disease	—	—
heart disease	—	0.826	heart disease	—	0.97
stroke	—	—	stroke	—	—
arthritis	—	—	arthritis	—	—
high cholesterol	0.396	0.361	high cholesterol	0.385	—
angina	0.525	0.359	angina	0.535	—
Measure	PA1	PA2	Measure	PA1	PA2
Proportion of Variance	0.145	0.109	Proportion of Variance	0.138	0.125

TABLE 4.3.4.4 Factor Loadings for Age ≥ 65 (left) and Age < 65 (right)

Again applying two filters viz. (i_mm=1 & I_residence=1) and (i_mm=1 & I_residence=0) we get,

Variable	PA1	PA2	Variable	PA1	PA2
high.bp (1 = no)	0.532	—	high.bp (1 = no)	0.58	—
diabetes (1 = no)	0.537	—	diabetes (1 = no)	0.622	—
cancer (1 = no)	—	—	cancer (1 = no)	—	—
lung disease	—	—	lung disease	—	—
heart disease	—	0.891	heart disease	—	0.842
stroke	—	—	stroke	—	—
arthritis	—	—	arthritis	—	—
high cholesterol	—	—	high cholesterol	0.402	—
angina	-0.487	0.334	angina	-0.489	0.333
Measure	PA1	PA2	Measure	PA1	PA2
Proportion of Variance	0.119	0.116	Proportion of Variance	0.134	0.103

TABLE 4.3.4.5 Factor Loadings for Urban Residents (left) and Rural Residents (right)

Again applying two filters viz. (i_mm=1 & I_occupation=1) and (i_mm=1 & I_occupation=0) we get,

Variable	PA1	PA2	Variable	PA1	PA2
high.bp (1 = no)	0.592	—	high.bp (1 = no)	0.544	—
diabetes (1 = no)	0.664	—	diabetes (1 = no)	0.587	—
cancer (1 = no)	—	—	cancer (1 = no)	—	—
lung disease	—	—	lung disease	—	—
heart disease	—	1.045	heart disease	—	0.848
stroke	—	—	stroke	—	—
arthritis	—	—	arthritis	—	—
high cholesterol	0.403	—	high cholesterol	0.351	—
angina	0.496	—	angina	-0.55	0.305
Measure	PA1	PA2	Measure	PA1	PA2
Proportion of Variance	0.147	0.141	Proportion of Variance	0.131	0.104

TABLE 4.3.4.6 Factor Loadings for Working Population (left) and Non-Working Population (right)

Again applying two filters viz. (i_mm=1 & I_alcohol=1) and (i_mm=1 & I_alcohol=0) we get,

Variable	PA1	PA2	Variable	PA1	PA2
high.bp (1 = no)	0.629	—	high.bp (1 = no)	0.563	—
diabetes (1 = no)	0.635	—	diabetes (1 = no)	0.606	—
cancer (1 = no)	—	—	cancer (1 = no)	—	—
lung disease	—	—	lung disease	—	—
heart disease	—	0.791	heart disease	—	0.892
stroke	0.313	—	stroke	—	—
arthritis	—	—	arthritis	—	—
high cholesterol	0.416	0.436	high cholesterol	0.371	—
angina	-0.55	0.348	angina	0.543	0.31
Measure	PA1	PA2	Measure	PA1	PA2
Proportion of Variance	0.162	0.109	Proportion of Variance	0.135	0.112

TABLE 4.3.4.7 Factor Loadings for Individuals consuming Alcohol (left) and Individuals not consuming Alcohol (right)

Again applying two filters viz. (i_mm=1 & I_exercise=1) and (i_mm=1 & I_exercise=0) we get,

Variable	PA1	PA2	Variable	PA1	PA2
high.bp (1 = no)	0.629	—	high.bp (1 = no)	0.569	—
diabetes (1 = no)	0.635	—	diabetes (1 = no)	0.642	—
cancer (1 = no)	—	—	cancer (1 = no)	—	—
lung disease	—	—	lung disease	—	—
heart disease	—	0.791	heart disease	—	0.962
stroke	0.313	—	stroke	—	—
arthritis	—	—	arthritis	—	—
high cholesterol	0.416	0.436	high cholesterol	0.394	—
angina	-0.55	0.348	angina	-0.547	0.303
Measure	PA1	PA2	Measure	PA1	PA2
Proportion of Variance	0.162	0.109	Proportion of Variance	0.143	0.125

TABLE 4.3.4.8 Factor Loadings for Individuals who exercise (left) and individuals who do not exercise (right)

Again applying two filters viz. (i_mm=1 & I_pension=1) and (i_mm=1 & I_pension=0) we get,

Variable	PA1	PA2	Variable	PA1	PA2
High BP	0.689	—	High BP	0.573	—
Diabetes	0.623	—	Diabetes	0.616	—
Cancer	—	1.058	Cancer	—	—
Lung Disease	—	—	Lung Disease	—	—
Heart Disease	—	—	Heart Disease	—	0.915
Stroke	0.357	—	Stroke	—	—
Arthritis	—	—	Arthritis	—	—
High Cholesterol	—	—	High Cholesterol	0.389	—
Angina	-0.556	—	Angina	-0.536	0.314
Measure	PA1	PA2	Measure	PA1	PA2
Proportion Var	0.162	0.147	Proportion Var	0.138	0.116

TABLE 4.3.4.9 Factor Loadings for Individuals who receive a pension (left) and individuals who do not receive a pension (right)

Again applying two filters viz. ($i_mm=1$ & $I_edu=1$) and ($i_mm=1$ & $I_edu=0$) we get,

Variable	PA1	PA2	Variable	PA1	PA2
High BP	0.609	—	High BP	0.575	—
Diabetes	0.582	—	Diabetes	0.6	—
Cancer	—	—	Cancer	—	—
Lung Disease	—	—	Lung Disease	—	—
Heart Disease	—	0.943	Heart Disease	—	0.814
Stroke	—	—	Stroke	—	—
Arthritis	—	—	Arthritis	—	—
High Cholesterol	—	—	High Cholesterol	0.396	—
Angina	-0.521	0.308	Angina	-0.525	0.354
Measure	PA1	PA2	Measure	PA1	PA2
Proportion Var	0.132	0.121	Proportion Var	0.135	0.102

TABLE 4.3.4.10 Factor Loadings for Individuals who are educated (left) and individuals who are not educated (right)

Applying two filters viz. ($i_mm=1$ & $I_srh=1$) and ($i_mm=1$ & $I_srh=0$) we get,

Variable	PA1	PA2	Variable	PA1	PA2
high.bp	0.578	—	high.bp	0.529	—
diabetes	0.627	—	diabetes	0.57	—
cancer	—	—	cancer	—	—
lung.disease	—	—	lung.disease	—	—
heart.disease	—	0.856	heart.disease	—	0.741
stroke	—	—	stroke	—	—
arthritis	—	—	arthritis	—	—
high.cholesterol	0.382	—	high.cholesterol	0.361	0.361
angina	—	—	angina	—	0.4
	0.563			0.611	
Metric	PA1	PA2	Metric	PA1	PA2
Proportion Var	0.143	0.101	Proportion Var	0.138	0.106

TABLE 4.3.4.11 Factor Loadings for Individuals who have $I_srh=1$ (left) and individuals who have $I_srh=0$ (right)

4.3.5 MULTIMORBIDITY INDEX (MMI):

Based on the results obtained from factor analysis, we created a multimorbidity index (MMI) based on the loadings derived from the two factors. These indices were computed as a weighted combination of various disease variables demonstrating the overall disease burden of the ageing population. These indices were compared across different socio-demographic and behavioural factors, and are summarised in the following table:

Independent Variables	0	1
Alcohol consumption	2.065	2.016
Exercise	2.055	2.06
Gender	2.001	2.144
Residence	1.933	2.016
Age	2.016	2.073
Education	1.95	2.134
Pension	2.046	2.077
Self-Reported Health	2.01	2.067

TABLE 4.3.5.1 Computation of MMI

We can conclude that the MMI was found to be higher among males, older individuals (age ≥ 65 years), urban residents, and those with higher education levels, and alcohol consumption showed minimal variation in the disease burden. This highlights the disparity across population subgroups while capturing the overall burden of the disease. The results are consistent with the patterns identified by correlation and factor analysis.

The following section talks about the findings that can be derived from the analysis that we have performed in the earlier sub-sections.

5. CONCLUSION:

- a. The descriptive analysis reveals that multimorbidity among the ageing population in India is largely concentrated at lower levels of disease burden, with most individuals experiencing zero, one, or two chronic conditions, while higher levels of multimorbidity are relatively found in lesser numbers. The distribution is positively skewed, indicating that only a small proportion of individuals suffer from multiple diseases simultaneously. From the analysis of age and multimorbidity (i_mm), we can

observe that the incidence of multimorbidity increases significantly in older age groups. Gender-wise comparisons suggest that females tend to exhibit slightly higher levels of multimorbidity, particularly at older ages. Furthermore, applying the chi-square test for association between the independent variables and i_mm confirms statistically significant associations between multimorbidity and key factors such as gender, alcohol consumption, and exercise, although the strength of these associations remains relatively weak, since the values of the contingency coefficient computed are very low (close to 1).

- b. Following the descriptive analysis, which provided the distribution and patterns of multimorbidity wrt different independent variables, regression analysis was undertaken to examine the underlying relationships more rigorously and to identify the key determinants influencing the likelihood of multimorbidity. From Regression Analysis, using Table 5.4.1, we can infer that the extent of multimorbidity is influenced by a combination of demographic, socio-economic, and health-related factors. Progression of age and pension received are linked with higher odds of multimorbidity. Also, it is interesting to note that as the financial condition improves, access to healthcare services also increases for individuals, thereby enhancing the incidence of multimorbidity. In contrast, variables such as gender (males) and occupation (working) reduced the likelihood of multimorbidity. The key reason for this may be that in our society, the earning male members of the family are given more care and importance, and hence they exhibit a lower level of multimorbidity as compared to women, who in the majority of the cases are non-working. From the MoSPI periodic labour force survey, the female workforce participation has been reported as 23.2% in 2017-18, which might be indicative of the fact that females might not be able to afford healthcare facilities.

Alcohol consumption and urban residence suggest a slightly higher risk of multimorbidity, as these variables show a moderate positive association with multimorbidity. The urban population have better access to healthcare facilities like hospitals, primary healthcare centers, etc leading to early detection of the disease and hence the higher incidence of multimorbidity as compared to the rural population. The most significant determinant factor that emerges is self-reported health, i.e. individuals reporting good health have lower odds of multimorbidity, the reason being that they assess themselves to be healthy individuals and hence they do not get their

health checkup done, leading to lower reporting of the number of diseases that they might suffer from. Education, family size and exercise exhibit limited influence on multimorbidity. Overall, we can conclude that health perception, ageing, and financial status are major determinants of multimorbidity.

- c. The significance of the estimated coefficients of logistic regression was further assessed through odds ratio (OR) testing. Based on the confidence intervals obtained in Table 5.4.1.1, we can conclude that most variables, except family size, are statistically significant predictors of multimorbidity. Health perception and age are the most influential determinants, while other socio-demographic covariates contribute only marginally.
- d. Factor analysis was conducted to explore the underlying patterns of multimorbidity wrt incidence of the diseases. The two prominent factors that emerged from the analysis were: **cardiometabolic risk cluster** and **cardiorespiratory risk cluster**, which account for 32.5% of the total variance. Here, heart disease appears to be the most prominent disease in both clusters. Further, to do an in-depth analysis of the factors in response to the binary levels of all of the independent variables, the factor analysis was done under the different filters applied on the data. The findings are reported in Table 5.4.2. Parallel Analysis Scree Plot support the idea that shared biological and lifestyle risk factors drive the co-occurrence of chronic conditions. Now, to further assess how different variables influence the factor formation, factor analysis was performed on the 9 binary categories of independent variables. The interpretations can be summarised as follows:
 - Factor Analysis of Disease Patterns by Gender: In both males and females, it can be observed that high blood pressure and diabetes exhibit strong loadings on Factor 1. Also, heart disease contributes the highest in the factor loadings observed in Factor 2 in both males and females, with the loadings being slightly stronger in males as compared to females.
 - Factor Analysis of Disease Patterns by Age: Similar to what was observed in gender analysis, high blood pressure and diabetes exhibit strong loadings on Factor 1, irrespective of the ages of the individuals. Heart disease dominates in the factor loadings of the 2nd factor.

- Factor Analysis of Disease Patterns by Residence: A consistent cardiometabolic disease cluster (high blood pressure and diabetes) exists irrespective of place of residence. Similar to earlier findings, here again, heart disease still dominates the factor loadings of Factor 2.
- Factor Analysis of Disease Patterns by Workforce: Cardiometabolic risks (High BP and Diabetes) dominate Factor 1 across both groups (working and non-working individuals). However, the factor loadings for these diseases were higher among working individuals. This might be potentially attributed to occupational stress, sedentary work environment and lifestyle related risk factors, which increase the cardiometabolic disease burden among the working population.
- Factor Analysis of Disease Patterns by Alcohol Consumption: Among individuals who consume alcohol, there was a higher loading in Factor 1 as compared to non-drinkers. This indicates that alcohol consumption is associated with an increased clustering of cardiometabolic risk factors.
- Factor Analysis of Disease Patterns by Physical Activity (Exercise): Diabetes and Heart disease had higher factor loadings in individuals who do not engage in regular physical activity. This reflects that physical activity is important in reducing cardiometabolic risk and the overall burden of multimorbidity.
- Factor Analysis of Disease Patterns by Pension: Factor analysis reveals a two factor structure across both groups with Factor 1 as cardio metabolic cluster and factor 2 as cardiovascular cluster. The individuals receiving a pension have higher factor loadings in comparison to those not receiving pension. This pattern may reflect better access to healthcare and diagnosis among pension recipients. Notably, cancer appears in factor 2 among pension receipts but not in other subgroup and overall analyses. This might indicate differential reporting instead of a true absence.
- Factor Analysis of Disease Patterns by Education: Similar to what was observed in the above group, high cholesterol contributes only in the non-educated group, suggesting that lifestyle-related risk factors may be more influential among individuals without education. Also, both factors exhibit higher loadings among educated individuals, which can be attributed to greater

health awareness, improved access to diagnostic services, and a higher likelihood of reporting conditions.

- Factor Analysis of Disease Patterns by Self-Reported Health: A consistent multimorbidity structure is observed across self-reported health groups, with hypertension, diabetes, and high cholesterol defining the cardiometabolic factor, and angina and heart-related conditions defining a cardiovascular factor.. Angina and heart-related conditions were present in the cardiovascular cluster. The strength of factor loadings was higher among individuals with good self-reported health, indicating a well-defined clustering of diseases.
- e. Having identified the underlying patterns and clustering of diseases through factor analysis and studying them across different independent variables, the multimorbidity index is now computed to quantify the overall disease burden of the aging population. Based on the MMI computed wrt the binary levels of the independent variables, we may conclude that:
- Although through regression analysis it was observed that females have higher odds of multimorbidity, we can now observe that males show a higher value of the MM Index as compared to females, suggesting that there is a greater disease severity and burden among multimorbid males.
 - A higher value of the MM Index among educated individuals might reflect that education helps individuals to be more aware of healthcare and diagnostic services, leading to a higher reporting of diseases in comparison to uneducated individuals.
 - Although it might be expected that rural residents have a higher disease burden, which contradicts the result obtained from the MM Index that urban residents exhibit a higher multimorbidity burden. This might be possible due to improved healthcare, including better awareness and access to diagnostic services, leading to early detection and a higher incidence of multimorbidity.
 - It is interesting to note that MMI is higher for the non-alcoholic population as compared to alcoholic population. Although, many health studies suggest that health statistic point to the fact that alcohol consumption leads to various diseases in human body but, the elevated levels of air pollution, particularly in urban areas, adulteration in food and water and sedentary

lifestyles are equally responsible for the occurrence of many diseases. Hence, the MMI in the absence of alcohol consumption is higher than in the presence of alcohol consumption.

In light of the above findings, it is important to derive relevant policy implications to address the growing burden of multimorbidity among the ageing population:

- The current healthcare system in India is largely focused on single-disease treatment, which is inadequate for managing multimorbidity. There is a need to transition healthcare models that simultaneously address multiple chronic conditions.
- The study also revealed that multimorbidity is closely linked to socioeconomic status, education, and residence (urban–rural divide), hence, policies must focus on reducing health inequalities by improving access to affordable healthcare services, especially for economically disadvantaged and rural populations.
- Special attention should be given to women, older age groups, low-income populations, and those with limited education, as they are more prone to multimorbidity, as confirmed by our regression analysis.
- Policies should focus on improving health literacy, particularly among less educated populations, through community-based awareness programs and preventive health campaigns.
- The strong clustering of cardiometabolic diseases emphasises the importance of lifestyle-based interventions, including promotion of physical activity, healthy diet, and reduction in substance.

6. LIMITATIONS OF THE STUDY:

- Since it is cross-sectional data, it is a snapshot of data. Hence, it is difficult to perform any time series analysis of our research problem. So, all our problems are based on the period 2017-2018.
- Most variables that we are studying are ordinal in nature, hence we cannot apply the parametric tools and techniques which confirm to the numeric data.
- A significant portion of the data is self-reported, which may introduce recall bias, social desirability bias, and potential underreporting of health conditions and lifestyle behaviours.

- The study primarily focuses on individuals aged 45 years and above, thereby restricting its applicability for a complete lifetime analysis.
- As an observational dataset, LASI also limits the ability to establish causal relationships.

7. FUTURE AND SCOPE OF THE STUDY:

We have used LASI data Wave-1, which was compiled in the year 2017-18, on the ageing population of India. With the availability of LASI Wave-2 data in future, we can examine the progression of accumulation patterns, multimorbidity incidence, survivorship functions, etc., keeping Wave-1 data as baseline. This will help in understanding the trajectory of multimorbidity in relation to the covariate factors and identify the various areas where improvement needs to be made as part of the policy in order to achieve a healthy elderly population. The study can be further extended to analyse the impact of multimorbidity on healthcare expenditure, mental health, quality of life, as well as impact of environmental and living 55 conditions, including early-life health and nutrition. It also provides scope to examine deeper socioeconomic and regional inequalities, along with disparities in terms of access to healthcare services and insurance coverage.

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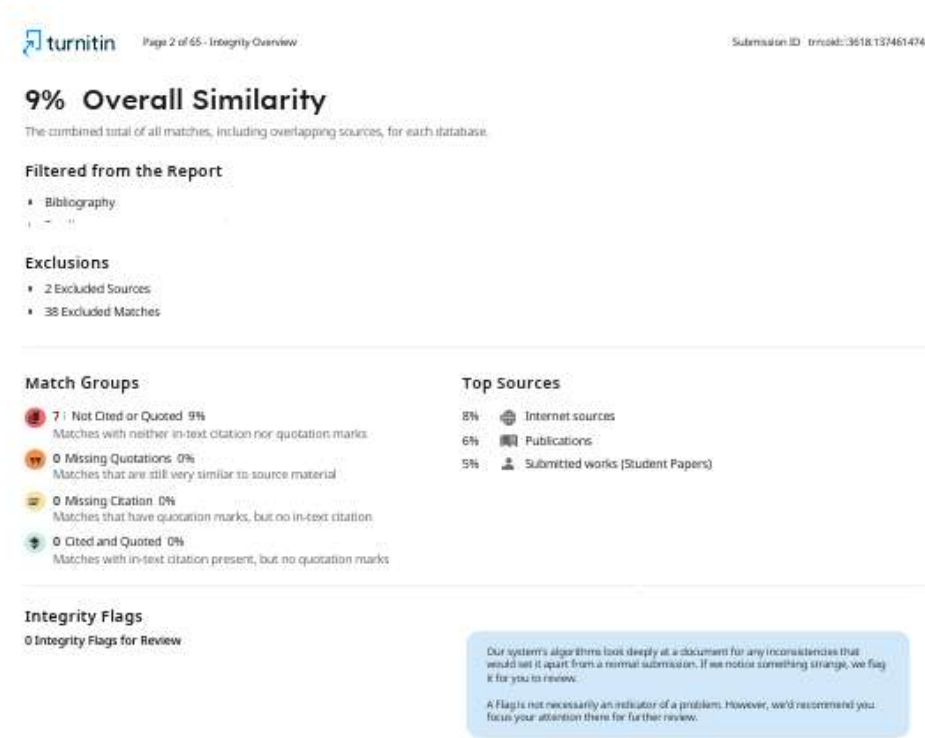
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9 APPENDIX I

I) Plagiarism report



ii) Ai report


Page 2 of 61 - AI Writing Overview

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10. APPENDIX-II

Participation Certificate for Presenting Paper in International Conference:

I presented the joint paper entitled “**Identification of Factors and their Impact on the multimorbidity pattern of aging population in India**” in the International Conference on Innovative Trends in Mathematics and Intelligent Systems (ITMIS) held in Shyam Lal College on 27th Feb, 2026. The certificate for the same is attached below:

